Status_Summary

Helix OTA — Feature Inventory — Status Summary

Revision: 5 **Last modified:** 2026-06-20T06:00:00Z **Companion of:** `Status.md` (Section 11.4.56 two-audience parity).

Page 1 — For the operator (plain language)

Helix OTA is an enterprise over-the-air update system. Here is what exists and what state it is in.

What works today:

- The **control-plane server** (Go/Gin) is fully built and tested — all APIs for managing devices, releases, deployments, rollouts, recalls, and audit logs have both unit tests and end-to-end tests running in containers. New endpoints: `GET /api/v1/devices` (device inventory listing) and `GET /devices/by-hardware/:hardwareId` (hardware-ID reverse lookup) both PASS.
- **Six reusable Go libraries** (protocol types, artifact validation, rollout engine, telemetry, HTTP/3) are built and tested with unit + stress + chaos tests.
- **Container infrastructure** is in place via the `vasic-digital/containers` submodule — tests run in podman pods.
- **The emulator ladder proves the A/B update core on this Mac without needing the real hardware:**
  - The container round-trip (control plane talking to a virtual device) works.
  - The emulated device boots to a live Linux userspace — full boot transcript captured as proof.
  - **The A/B slot switch is proven** — a real U-Boot bootloader switches between system slot A and slot B on demand, with 3/3 identical passes.
  - **Automatic rollback on a corrupt slot is proven** — when we mark a slot bad, the bootloader detects it and falls back to the good slot automatically.
  - **RAUC dm-verity slot integrity (PWU-AB-2) is proven** — direct-dd A/B slot switch GREEN 3/3 deterministic with captured evidence.
  - **ApplyPort (PWU-AB-4) is implemented** — slot manager, ed25519 signature verifier, health marker, HTTP client, and CLI binary — 36 tests, 3 Go source files, 2 Kotlin files, all tested and passing.
- **Security tests**, **e2e tests**, and **pre-build gates** all pass.
- **Production deployment** is live on `nezha.local` — a full 3-container stack (server, PostgreSQL, SPA) orchestrated via docker-compose.

- **Remote stress testing** proves 291 req/s sustained device registration with 100/100 devices and all stress/chaos tests passing.
- **Security hardening** — IDOR project-scoped authorization prevents cross-project access; Tauri IPC scoped to minimum permissions; docker-compose secrets no longer ship default credentials.
- **MountManagerUI** bug fixed — the SPA now serves correctly at `/manager/`.
- **Governance** (Issues, Fixed, CONTINUATION, README, Status docs) is maintained and in sync.

What is still pending (NOT done yet — honestly):

- **Android OTA agent on-device verification** — the two Kotlin modules exist and the ApplyPort Go/Kotlin code is implemented but has not been tested against a real Android target.
- **Full Android OTA test (Tier-2 Cuttlefish)** — cannot run on this Mac (needs Linux with KVM). Ready for the operator's Linux machine.
- **Real hardware testing (Tier-3 RK3588 board)** — no board on the bench yet.
- **Stress/chaos coverage** — present for some submodules but not yet for the server endpoints.
- **Build resource statistics tracking** — mandated but not yet built.
- **Workable-items SQLite database** — mandated but not yet built.
- **CodeGraph MCP integration** — completed (31,718 nodes indexed across own-org submodules).

Bottom line: The server, Go libraries, remote deployment, remote stress testing, the A/B update core (slot switch + auto-rollback + RAUC dm-verity), and ApplyPort server-side implementation are all proven with captured evidence or testing. Security hardening (IDOR, Tauri IPC, docker-compose secrets) is complete. The Android on-device apply, and hardware-tier tests remain as the open items before a release tag.

Page 2 — For software engineers

Feature inventory summary (all 107 items from Status.md):

Status	Count	Key Items
PASS	50	All server handlers (F01-F34), emulator Tier-0/Tier-1 (F44-F49), e2e tests (F57-F67), build gates (F69-F71), scripts (F74-F76), Multi-Project API + IDOR (F90-F91), MountManagerUI (F98), IDOR Security (F99), Tauri IPC (F100), Docker Secrets (F101), Remote Deploy (F103), Devices List API (F104), Hardware ID Reverse

Status	Count	Key Items
		Lookup (F105), Demo Re-recordings (F107)
VERIFIED	20	Go submodules (F35-F41), containers (F68), .gitignore (F73), governance (F77-F85), CodeGraph wired (F88), frontend build + tests (F92-F93), Production Deploy (F96), Remote Stress (F97), §11.4.159 Recording Compliance (F106)
PROVEN	6	PWU-AB-1 base+boot (F50), slot switch (F51), auto-rollback (F52), PWU-AB-2 RAUC dm-verity (F53), slot switch video (F94), rollback video (F95)
IMPLEMENTED	2	PWU-AB-4 ApplyPort (F54), ApplyPort Scaffold (F102)
DESIGN	2	ota-android-agent (F42), ota-update-engine-bridge (F43)
OPERATOR-BLOCKED	2	Tier-2 Cuttlefish (F55 — needs Linux+KVM), Tier-3 HW (F56 — needs board)
PARTIAL	2	Stress+chaos coverage (F86 — 2/12 submodules), Docs Chain (F89 — engine built, not submoduled)
NOT_STARTED	2	Build-resource-stats (F72), workable-items DB (F87)

Proven A/B core (captured evidence): - **PWU-AB-1 slot switch** — `docs/qa/20260611T094958Z-ab-slot-switch/` (3/3 deterministic PASS, real U-Boot 2024.01 on QEMU virt + HVF) + MP4 recording (1.3 MB) - **PWU-AB-3 auto-rollback** — `docs/qa/20260611T095918Z-ab-rollback/` (bad slot -> bootcount exceeded -> altbootcmd swap -> known-good slot; CONTROL proves rollback only fires on bad slot) + MP4 recording (1.4 MB) - **PWU-AB-2 RAUC dm-verity** — `docs/qa/20260620T051026Z-ab-rauc-verity/` (direct-dd A/B slot switch, 3/3 deterministic, dd clone rc=0, fw_setenv rc=0, post-slot HELIX_POSTSLOT=B confirmed) - **Video recordings** — All content-verified per §11.4.158 liveness battery.

Pending (honest per Section 11.4.6): - PWU-AB-2 RAUC dm-verity: **PROVEN — GREEN 3/3 deterministic** via direct-dd - PWU-AB-4 ApplyPort: **IMPLEMENTED** — slot manager, signature verifier, health marker, HTTP client, CLI binary — 36 tests, 3 Go + 2 Kotlin files; NOT yet tested against a real Android target - Tier-2: `tier2_cuttlefish_ab.sh` authored, SKIPs on this host (no `/dev/kvm`) - Tier-3: No physical board

No release tag — Section 11.4.40 requires full ladder GREEN; Tier-2 + Tier-3 remain blocked.

Video Recording Gaps

Priority	Gap	Depends On
High	Tier-2 Cuttlefish Android OTA apply screen recording	Linux+KVM host
High	Tier-3 RK3588 HDMI capture of on-device OTA	Physical board
Medium	Tier-1 AVD + HVF screen recording	Existing qa evidence may be partial
Low	Tier-0 container round-trip	Console logs suffice
Fixed	A/B slot switch + rollback	2 MP4 recordings captured (1.3 MB + 1.4 MB), content-verified per §11.4.158
Future	Web UI screen recording	Web UI not yet built
N/A	Audio routing	No audio subsystem in scope

Section-refs

Section 11.4.45 (Status doc), Section 11.4.56 (two-audience summary), Section 11.4.5 (captured-evidence table), Section 11.4.6 (no-guessing), Section 11.4.44 (revision header), Section 11.4.65 (universal export).